

4.1 Double Integrals over Rectangular Regions Worksheet

1. Find the volume of the solid beneath the paraboloid $f(x, y) = 12 - x^2 - 2y^2$ and above the region $R = \{(x, y) \mid 1 \leq x \leq 2, 0 \leq y \leq 1\}$.

A picture of the region R is shown at the right.

We know that the volume of the solid is given by

$$V = \int_0^1 \int_1^2 (12 - x^2 - 2y^2) dx dy$$

We calculate this double integral as follows:

$$\begin{aligned} V &= \int_0^1 \int_1^2 (12 - x^2 - 2y^2) dx dy = \int_0^1 \left(12x - \frac{1}{3}x^3 - 2y^2x \right) \Big|_1^2 dy = \int_0^1 \left[\left(24 - \frac{8}{3} - 4y^2 \right) - \left(12 - \frac{1}{3} - 2y^2 \right) \right] dy \\ &= \int_0^1 \left[\frac{29}{3} - 2y^2 \right] dy = \left[\frac{29}{3}y - \frac{2}{3}y^3 \right] \Big|_0^1 = \frac{29}{3} - \frac{2}{3} = \frac{27}{3} = 9 \end{aligned}$$



2. Calculate $\int_0^1 \int_1^2 \frac{xe^{x^2}}{2y} dy dx$.

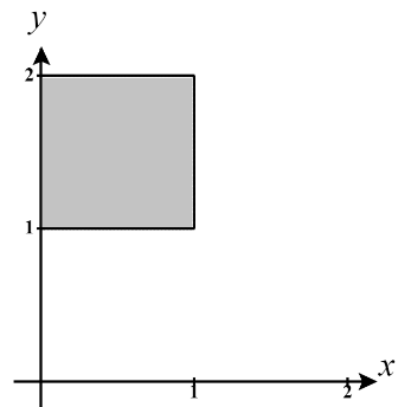
A picture of the region of integration is shown at the right.

We calculate this double integral as follows:

$$\begin{aligned} V &= \int_0^1 \int_1^2 \frac{xe^{x^2}}{2y} dy dx = \int_0^1 \frac{1}{2} xe^{x^2} \ln(2y) \Big|_1^2 dx \\ &= \int_0^1 \left[\frac{1}{2} xe^{x^2} \ln(4) - \frac{1}{2} xe^{x^2} \ln(2) \right] dx \\ &= \int_0^1 \frac{1}{2} xe^{x^2} \ln(2) dx \end{aligned}$$

Let $u = x^2$ then $du = 2xdx \rightarrow \frac{1}{2} du = xdx$. Our bounds become $u(0) = 0$ & $u(1) = 1$. Then,

$$\int_0^1 \frac{1}{2} xe^{x^2} \ln(2) dx = \frac{1}{4} \ln(2) \int_0^1 e^u du = \frac{1}{4} \ln(2) e^u \Big|_0^1 = \frac{1}{4} \ln(2) [e^1 - e^0] = \frac{1}{4} \ln(2) [e - 1]$$



3. Determine the value of the volume communicated by the double integral $\iint_D 3 \, dA$ for each of the following regions, using only simple geometry formulas.

a) $D = \{(x, y) \mid 1 \leq x \leq 2 \text{ \& } 2 \leq y \leq 10\}$.

We are integrating over a rectangle here with dimensions of 1 and 8. We are finding the volume below the surface of $z = 3$ and so our region is the shape of a rectangular prism with height 3. Therefore,

$$V = \iint_D 3 \, dA = (1)(8)(3) = 24.$$

b) D is a circle of radius 4.

We are integrating over a circle and finding the volume below the surface of $z = 3$ and so our region is the shape of a cylinder. Therefore, $V = \iint_D 3 \, dA = \pi(4)^2(3) = 48\pi$.

c) D is determined by the curves, $x = 0$, $y = 0$, & $y = 1 - x$.

We are integrating over a triangle region with base 1 and height 1. This means that we are trying to find the volume of a triangular prism. Therefore, $V = \iint_D 3 \, dA = \frac{1}{2}(1)(1)(3) = \frac{3}{2}$.

4. Determine if each of the following statements are TRUE or FALSE.

a) Suppose that $f(x, y)$ is a continuous function over $\mathbb{R}^2$. There is no difference in the values of the integrals $\int_{-1}^5 \int_{-1}^1 f(x, y) \, dx \, dy$ and $\int_{-1}^1 \int_{-1}^5 f(x, y) \, dy \, dx$.

TRUE! The region being integrated over here is rectangular and so the order of integration can always be changed.

b) A double integral can be used to find volume under a surface and also the area of a region.

TRUE! Remember that $\iint_D 1 \, dA$ represents the volume of the rectangular prism with base area of D and height of 1. Therefore $V = (1)(D) = D$. This means that the double integral can compute a volume but since it returns the area of region D as this volume, we can also think of this double integral as computing the area of D .

c) The double integral $\iint_D 3 \, dA$ represents triple the area of D .

TRUE! Note that $\iint_D 3 \, dA = 3 \iint_D 1 \, dA$ and we know that $\iint_D 1 \, dA$ represents can be thought to represent the area of D .